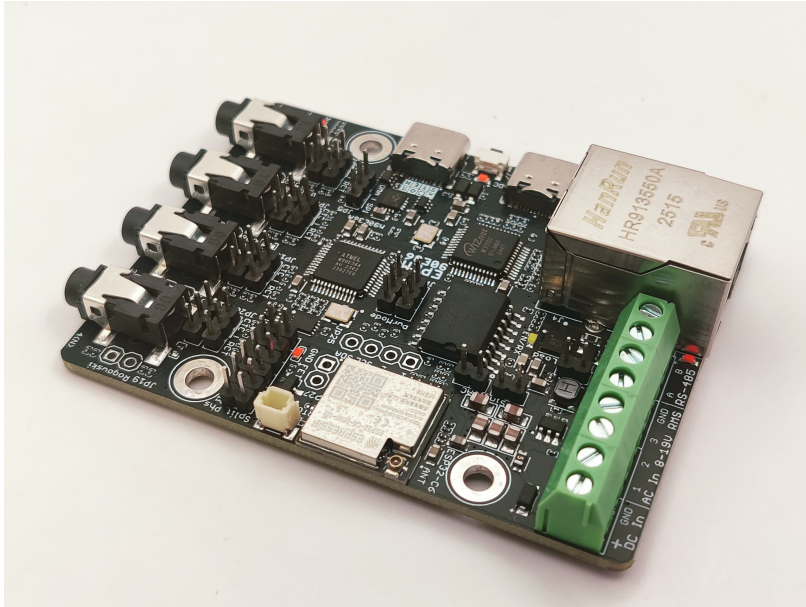


DTM-EMS

Precision Distribution-Transformer Monitoring with Neutral, Phase & Leakage Insight



DTM-EMS pairs a proven off-the-shelf precision-metering platform ($\pm 0.1\%$ polyphase) with an open-source insights layer — delivering transformer-grade monitoring of neutral, phase and leakage, plus dynamic energy-policy enforcement at the secondary distribution edge.

What Sets DTM-EMS Apart:

- ▶ **Polyphase Precision Metering** : $\pm 0.1\%$ active-energy accuracy, 3-phase 4-wire with a dedicated neutral channel.
- ▶ **Neutral / Phase / Leakage Insight** : measures neutral current and phase imbalance directly, and supports non-intrusive load monitoring (NILM).
- ▶ **Open Edge Integration** : an expansion HAT exposes the meter over SPI / 40-pin header to the NESL co-processor for dynamic policy enforcement (stackable up to 64 boards).

DTM-EMS gives operators a precise, intelligent view of the secondary transformer — neutral, phases and leakage — and the means to enforce energy policy at the edge.

Intelligent

- ▶ Neutral / phase / leakage + NILM insights
- ▶ Polyphase metering, $\pm 0.1\%$ active energy
- ▶ Dynamic edge policy enforcement

Interoperable

- ▶ Ethernet + RS-485 Modbus
- ▶ SPI / 40-pin header to any MCU / Pi
- ▶ WiFi 6 / Thread / Zigbee

Reliable

- ▶ Proven off-the-shelf metering hardware
- ▶ Onboard RTC, temperature & EEPROM
- ▶ Stackable up to 64 boards

PRODUCT	
Product name	DTM-EMS (10Power X-EMS, supplied by NESL)
Metering platform	Off-the-shelf polyphase precision-metering platform
Firmware	meshEMS-integrated open-source insights & policy layer
METROLOGY	
Metering IC	Polyphase energy-metering IC
Channels	3 voltage inputs up to 4 CT: 3 phase + 1 neutral (clamp or Rogowski)
Accuracy	$\pm 0.1\%$ active energy $\pm 0.2\%$ reactive energy
INSIGHTS	
Insights engine	Neutral phase imbalance leakage NILM (non-intrusive load monitoring)
Measured	RMS V / I P / Q / S harmonics PF phase angle frequency bidirectional energy
POLICY ENFORCEMENT	
Edge enforcement	Dynamic energy-policy enforcement at the transformer edge
Coordination	Site-networked EMS + DER coordination
INTEGRATION & NETWORKING	
Co-processor link	SPI + 40-pin expansion header (any MCU / Pi, stackable $\times 64$)
Networking	Ethernet (RJ45) RS-485 Modbus WiFi 6 / Thread / Zigbee
Onboard	RTC temperature sensor EEPROM RGB status
POWER & MECHANICAL	
Power supply	5 V USB-C or 4.5–20 V DC or 8–12 V AC (SELV)
Dimensions	85 $\times$ 55 mm or Raspberry Pi HAT form factor
STANDARDS & POSITIONING	
Grid position	Secondary distribution transformer (DTM)
Capabilities	Polyphase precision metering neutral / phase / leakage insight edge policy enforcement

DTM-EMS combines an off-the-shelf precision-metering platform with an open-source insights and policy layer by New Energy Solutions Lab for 10Power. Customizable with added modules and open or licensed code stacks; items marked **[to confirm]** to be confirmed.